2.28 Numerical Methods: $x = \mathrm{g}(x)$ Method

Worked Solutions 1–18

1 a)

$x^2 = 4 - x \Rightarrow x = \sqrt{4 - x}$ as req.

1 b)

$x^2 + x - 4 = x(x + 1) - 4 = 0 \Rightarrow x(x + 1) = 4 \Rightarrow x = \frac{4}{x+1}$ as req.

2 a)

$x^3 = 5 - 2x \Rightarrow x = \sqrt[3]{5 - 2x}$ as req.

2 b)

$x^3 + 2x - 5 = x(x^2 + 2) - 5 = 0 \Rightarrow x(x^2 + 2) = 5 \Rightarrow x = \frac{5}{x^2+2}$ as req.

3 a)

$$x^3 - x^2 - 2x - 3 = x^2(x - 1) - 2x - 3 = 0$$

$$x^2(x - 1) = 3 + 2x \Rightarrow x^2 = \frac{3 + 2x}{x - 1} \Rightarrow x = \sqrt{\frac{3 + 2x}{x - 1}} \text{ as req.}$$

3 b)

$$x_{n+1} = \sqrt{\frac{3 - 2x_n}{x_n - 1}}$$

$x_1 = 2$, $x_2 = 2.645751311\ldots$, $x_3 = 2.2445777041\ldots$, ... Solution is $x = 2.3744$ to 4 decimal places.

4 a)

$\mathrm{f}(3) = -5.112489402\ldots < 0$, $\mathrm{f}(4) = 2.180709778\ldots > 0$. Since $\mathrm{f}(x)$ is continuous and there is a change of sign then there is a solution between $x = 3$ and $x = 4$.

4 b)

$x^2 \ln x = 5x \Rightarrow x^2 = \frac{5x}{\ln x} \Rightarrow x = \sqrt{\frac{5x}{\ln x}}$ as req.

4 c)

$$x_{n+1} = \sqrt{\frac{5x_n}{\ln(x_n)}}$$

$x_1 = 3$, $x_2 = 3.695076237\ldots$, $x_3 = 3.759747791\ldots$, ... Solution is $x = 3.7687$ to 4 decimal places.

5 a)

$x_2 = 2.8284$, $x_3 = 2.4142$, $x_4 = 2.3268$

5 b)

$x_2 = 1.6829$, $x_3 = 1.9874$, $x_4 = 1.8289$

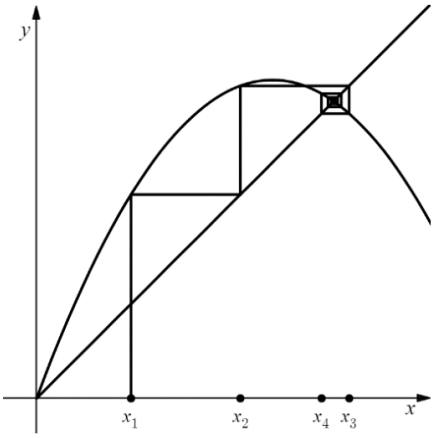
5 c)

$x_2 = 7.9000$, $x_3 = 5.6429$, $x_4 = 7.0444$

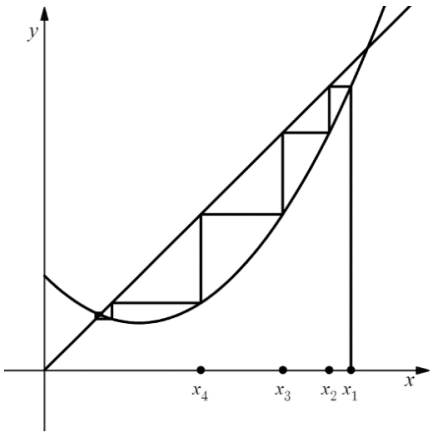
5 d)

$x_2 = 1.8000$, $x_3 = 1.6832$, $x_4 = 1.6353$

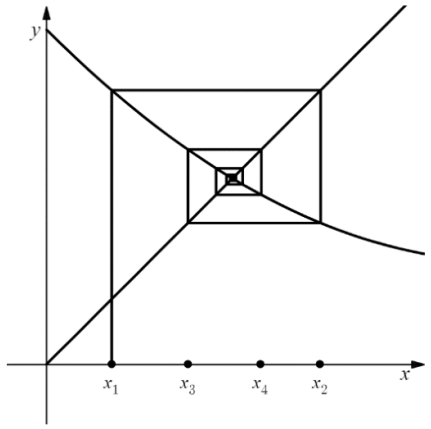
6 a)



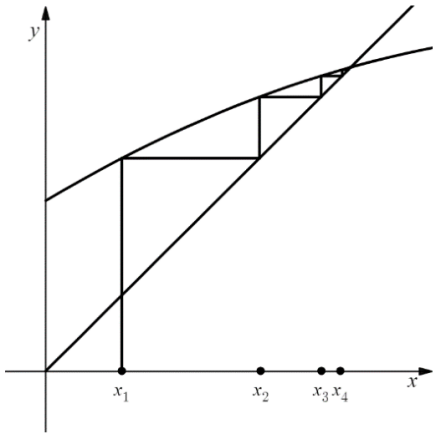
6 b)



6 c)



6 d)



7 a)

It will staircase and then cobweb to the intersection at $x = \frac{1}{4}$

7 b)

It directly gives the intersection at $x = \frac{1}{4}$

7 c)

It will cobweb or staircase/cobweb to the intersection at $x = \frac{1}{4}$

7 d)

It directly gives the intersection at $x = 1$

7 e)

It diverges away (as a staircase) from the solution at $x = 1$

8 a)

n	x_n
2	3.098612289...
3	3.130954362...
4	3.141337866...
5	3.144648781...
6	3.145702209...
7	3.146037143...
8	3.146143611...

Therefore $\alpha \in [3.146, 3.147]$

8 b)

n	x_n
2	1.16861989...
3	1.154479721...
4	1.163534853...
5	1.156892261...
6	1.161754207...
7	1.158189672...
8	1.160799868...
9	1.158886807...
10	1.160288018...
11	1.159261221...
12	1.160013389...
13	1.159462256...

Therefore $\alpha \in [1.159, 1.160]$

8 c)

n	x_n
2	1.4
3	1.71584
4	1.706983453...
5	1.711456501...
6	1.709231821...
7	1.710347038...
8	1.709790159...
9	1.71006878...

Therefore $\alpha \in [1.709, 1.710]$

9 a)

$g(x) = x^5 - 2x^4 - 5$, $g'(x) = 5x^4 - 8x^3$.

$$g'(2.273) = 5(2.273)^4 - 8(2.273)^3 = 39.51 > 1$$

So it diverges from the root.

9 b)

$\mathrm{g}(x) = e^x - 2$, $\mathrm{g}'(x) = e^x$. $\mathrm{g}'(1.146) = e^{1.146} = 3.146 > 1$ so it diverges from the root.

9 c)

$\mathrm{g}(x) = e^x - 2$, $\mathrm{g}'(x) = e^x$. $\mathrm{g}'(-1.841) = e^{-1.841} = 0.1587 < 1$ so it converges to the root.

10 a)

Iterative formula B: $x_2 = 2.815559633\ldots$, $x_3 = 2.827187864\ldots$, $x_4 = 2.835855288\ldots$, …. $x = 2.86$ (2 DP).

Iterative formula C: $x_2 = 1.296296296\ldots$, $x_3 = 1.224223602\ldots$, $x_4 = 1.276836976\ldots$, …. $x = 1.25$ (2 DP).

Iterative formula D: $x_2 = -1.154700538\ldots$, $x_3 = -1.097740076\ldots$, $x_4 = -1.121743696\ldots$, …. $x = -1.11$ (2 DP).

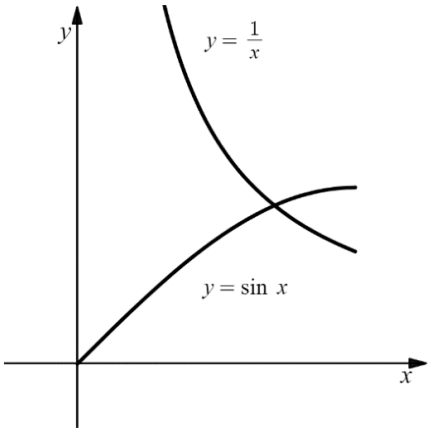
10 b)

$\mathrm{g}(x) = x^3 - 3x^2 - x + 4$, $\mathrm{g}'(x) = 3x^2 - 6x - 1$.

$$\mathrm{g}'(2.86) = 6.3788 > 1, \quad \mathrm{g}'(1.25) = -3.8125 < -1, \quad \mathrm{g}'(-1.11) = 9.3563 > 1$$

Convergence requires $-1 < \mathrm{g}'(\alpha) < 1$ which does not hold for any of the roots.

11 a)



11 b)

$\mathrm{f}(x) = \frac{1}{x} - \sin x$.

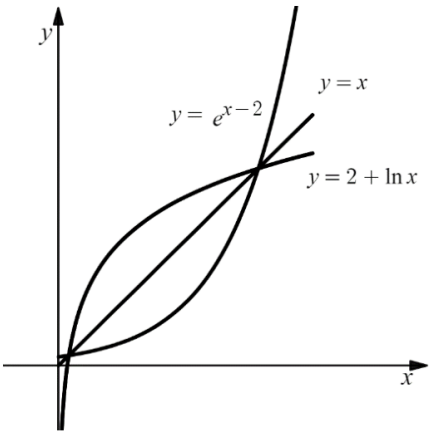
$$\mathrm{f}(1) = \tfrac{1}{1} - \sin 1 = 0.1585 > 0$$
$$\mathrm{f}(1.2) = \tfrac{1}{1} - \sin 1 = -0.0987 < 0$$

$\mathrm{f}(x)$ is continuous and there is a change of sign between $x = 1$ and $x = 1.2$ therefore there is a solution to $\frac{1}{x} = \sin x$ between $x = 1$ and $x = 1.2$.

11 c)

$x_{n+1} = \frac{1}{\sin(x_n)}$. $x_1 = 1$, $x_2 = 1.188395106\ldots$, $x_3 = 1.07785184\ldots$, $x_4 = 1.35146857\ldots$, …. The solution is $x = 1.1142$ to (4 DP).

12 a)



12 b)

All3 graphs intercept at the same points.

13 a)

$0 = 3T - e^{0.1T} \Rightarrow 0.1T = \ln(3T) \Rightarrow T = 10\ln(3T)$ as req.

13 b)

$T_{n+1} = 10\ln(3T_n)$. $T_2 = 49.698133\ldots$, $T_3 = 50.04579656\ldots$, $T_3 = 50.11550806\ldots$, …. $T = 50.133$ hours.

13 c)

50 hours

14 a)

Segment: $\frac{1}{2}r^2\theta - \frac{r^2}{2}\sin\theta$

$$\frac{1}{6}\pi r^2 = \frac{1}{2}r^2\theta - \frac{r^2}{2}\sin\theta \Rightarrow \frac{1}{6}\pi = \frac{1}{2}\theta - \frac{1}{2}\sin\theta \Rightarrow \frac{\pi}{3} = \theta - \sin\theta \text{ as req.}$$

14 b)

$$\frac{\pi}{3} = \theta - \sin\theta \Rightarrow \theta = \sin\theta + \frac{\pi}{3}$$

$\theta_{n+1} = \sin\theta_n + \frac{\pi}{3}$. $\theta_1 = 2$, $\theta_2 = 1.956494978\ldots$, $\theta_3 = 1.9737333373\ldots$, $\theta_4 = 1.967110835\ldots$, …. $\theta = 1.97$ radians (2 DP).

15 a)

$x = \cos x$, $x_{n+1} = \cos(x_n)$. $x_1 = 0.5$, $x_2 = 0.8775825619\ldots$, $x_3 = 0.6390124942\ldots$, $x_4 = 0.8026851007\ldots$, …. $x = 0.739$ (3 DP).

15 b)

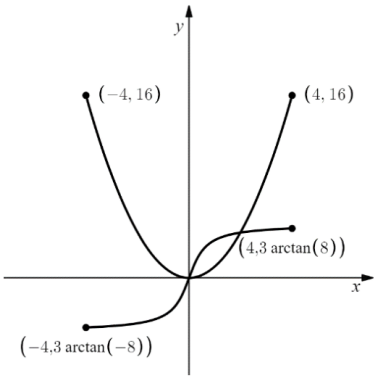
$$x^2 = x\log_3(0.5x+1) \Rightarrow x^2 - x\log_3(0.5x+1) = 0 \Rightarrow x(x - \log_3(0.5x+1)) = 0$$

Therefore $x = 0$. $x - \log_3(0.5x + 1) = 0 \Rightarrow x = \log_3(0.5x + 1)$. $x_{n+1} = \log_3(0.5x_n + 1)$ DIVERGES.

$$3^x = 0.5x + 1 \Rightarrow 3^x - 1 = 0.5x \Rightarrow x = 2 \times 3^x - 2$$

$x_1 = -1$, $x_2 = -1.333333333\ldots$, $x_3 = -1.53775915\ldots$, $x_4 = -1.630739868\ldots$, …. $x = -1.686$ (3 DP).

16 a)



16 b)

$\mathrm{f}(x) = x^2 - 3\arctan(2x)$.

$$\mathrm{f}(1.8) = 1.8^2 - 3\arctan(3.6) = -0.6595\ldots < 0$$

$$\mathrm{f}(2.2) = 2.2^2 - 3\arctan(4.4) = 0.7980\ldots > 0$$

Since $\mathrm{f}(x)$ is continuous and there is a change in sign between $x = 1.8$ and $x = 2.2$ there must be a root.

16 c)

$x_{n+1} = \sqrt{3\arctan(2x_n)}$. $x_1 = 2$, $x_2 = 1.994355282\ldots$, $x_3 = 1.993854416\ldots$, $x_4 = 1.993809838\ldots$, $x_5 = 1.99380587\ldots$. $x = 1.994$ (3 DP).

17 a)

$$y = -\frac{2x^2 + 24}{3\sqrt{x}} - 4\ln x = -\tfrac{2}{3}x^{3/2} - 8x^{-1/2} - 4\ln x$$

$$\frac{dy}{dx} = -x^{1/2} + 4x^{-3/2} - \frac{4}{x} = -\sqrt{x} + \frac{4}{x\sqrt{x}} - \frac{4}{x}$$

Stat pt when $\frac{dy}{dx} = 0$.

$$-\sqrt{x} + \frac{4}{x\sqrt{x}} - \frac{4}{x} = 0 \Rightarrow -x^2 + 4 - 4\sqrt{x} = 0$$

$$-x^2\sqrt{x} + 4\sqrt{x} - 4x = 0 \Rightarrow -\tfrac{1}{4}x^2\sqrt{x} + \sqrt{x} - x = 0 \text{ as req.}$$

17 b)

$x_{n+1} = \sqrt{x_n} - \frac{1}{4}x_n^2\sqrt{x_n}$. $x_1 = 1.5$, $x_2 = 0.75$, $x_3 = 0.7442405814\ldots$, $x_4 = 0.7432335552\ldots$, …. $x = 0.74$ (2 DP).

18 a)

$x_2 = 0.3165$ (4 DP), $x_3 = 0.3122$ (4 DP), $x_4 = 0.3133$ (4 DP)

18 b)

$$x = \tfrac{1}{4}\arccos x \Rightarrow 4x = \arccos x \Rightarrow \cos(4x) = x$$

Use double angle formula:

$$x = 1 - 2\sin^2(2x)$$

Use of double angle formula:

$$x = 1 - 2(2\sin x \cos x)^2 \Rightarrow x = 1 - 8\sin^2 x \cos^2 x$$

Therefore $x - 1 + 8\sin^2 x \cos^2 x = 0$ as req.